

1 Datasets structure

Table 1 summarizes all the datasets, their content, and how they are used through SARVI training and evaluation.

Table 1: Dataset usage throughout SARVI.

Task	CT-EBM-SP	E3C	CodiEsp	Code Dictionary	CARES	SpCN-Synth	Med LexSp
NER							
DISO	✓	✓	✓	□	□	■	▲
LIVB	✓	✓	□	□	□	□	▲
Date	✓	✓	□	□	□	□	▲
Neg_cue	✓	□	□	□	□	□	▲
Spec_cue	✓	□	□	□	□	□	▲
RE							
DISO-Date	✓	✓	□	□	□	□	△
DISO-LIVB	✓	□	□	□	□	□	△
DISO-Neg_cue	✓	□	□	□	□	□	△
DISO-Spec_cue	✓	□	□	□	□	□	△
ATT							
DISO	✓	□	□	□	□	□	△
ICD-10							
ICD-10-CM	□	□	✓	✓	■	■	▲
Judger	□	□	□	✓	□	□	△

Legend. ✓ Present in dataset; □ Not present in dataset; ■ Only used for evaluation; ▲ Used as support information.

The following figures represent the ICD-10-CM coverage of each of the datasets used. Each sector represents the total number of codes in the dataset that begin with the corresponding letter, while the filled proportion indicates the coverage of ICD-10-CM codes within that letter category. Fig. 1 shows the codes distribution in the CodiEsp training set, the SpCN-Synth one is plotted in Fig. 3, and finally, the distribution of the CARES test set is presented in Fig. 2.

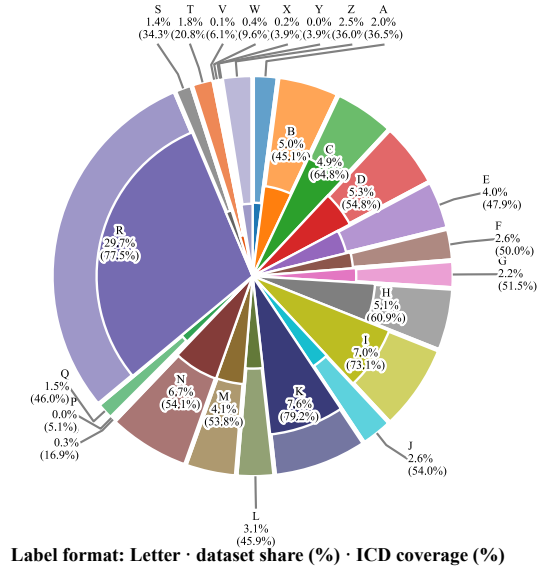
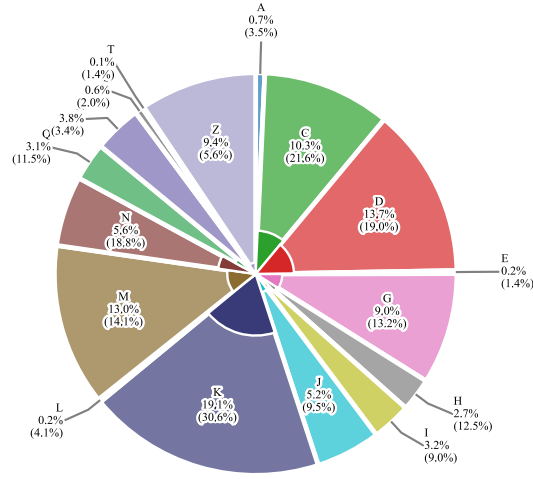
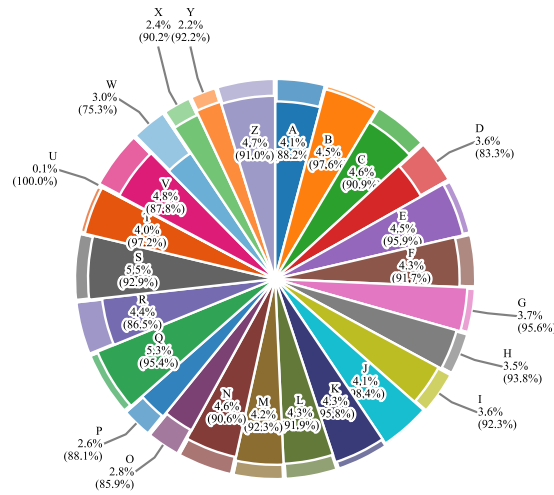


Figure 1: ICD-10-CM distribution - CodiEsp training set. A total of 809 unique category-level codes are represented. Also, 1143 detailed codes exist in the dataset.



Label format: Letter · dataset share (%) · ICD coverage (%)

Figure 2: ICD-10-CM distribution - CARES test set. A total of 143 unique category-level codes are represented. Also, 206 detailed codes exist in the dataset.



Label format: Letter · dataset share (%) · ICD coverage (%)

Figure 3: ICD-10-CM distribution - SpCN-Synth dataset. A total of 1746 unique category-level codes are represented. Also, there are no detailed codes in the dataset.

Fig. 4 displays a plot of the distribution of how many definitions per code there are. All codes are grouped and counted regarding the number of definitions that have been created for each one. Most codes are associated with relatively few definitions, whereas only a small proportion have many definitions, and a maximum of 200 definitions is retained per code.

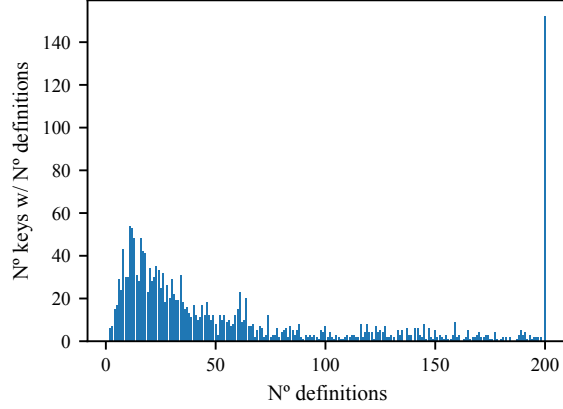


Figure 4: Number of definitions associated with each code.

2 NER results

The following Tables 2, 3, and 4 displays the metrics obtained in the NER step of the first stage of SARVI. Results obtained for all the entities are quite low focusing in F1 score. However, type errors reveal a big quantity of type 1 and type 5 failures, leading to only having false positive and correct entities but overlapping spans. As stated in the main paper, these errors can be principally made by the multiple medical domains that exist and how each expert annotates each entity throughout their datasets. False positives are maybe entirely correct entities with their corresponding ICD-10-CM code but are not annotated in the text. It is especially noticeable that in the SpCN-Synth dataset where there are no type 2 or type 4 errors.

Entities [Support]	Method	P	R	F1	Type 1	Type 2	Type 3	Type 4	Type 5	Total Errors
DISO										
3665	BIO	<u>0.385</u>	0.574	<u>0.461</u>	1918 [55.15 %]	3 [0.09 %]	87 [2.50 %]	22 [0.63 %]	1448 [41.63 %]	3478
	Span	0.424	0.527	0.470	1041 [37.50 %]	10 [0.36 %]	73 [2.63 %]	68 [2.45 %]	1584 [57.06 %]	2776
	Mix	<u>0.385</u>	<u>0.570</u>	0.460	1882 [54.42 %]	3 [0.09 %]	92 [2.66 %]	27 [0.78 %]	1454 [42.05 %]	<u>3458</u>

Table 2: NER: CodiEsp [SARVI stage 1: Post-training results]

Entities [Support]	Method	P	R	F1	Type 1	Type 2	Type 3	Type 4	Type 5	Total Errors
DISO										
251	BIO	0.350	0.398	0.372	43 [22.16 %]	5 [2.58 %]	1 [0.52 %]	12 [6.19 %]	133 [68.56 %]	194
	Span	0.007	0.124	0.013	4296 [95.13 %]	0	4 [0.09 %]	0	216 [4.78 %]	4516
	Mix	0.350	0.398	0.372	43 [22.16 %]	5 [2.58 %]	1 [0.52 %]	12 [6.19 %]	133 [68.56 %]	194
LIVB										
62	BIO	0.030	0.032	0.031	6 [9.09 %]	3 [4.55 %]	0	1 [1.52 %]	56 [84.85 %]	66
	Span	0.007	0.113	0.012	1004 [94.81 %]	0	1 [0.09 %]	0	54 [5.10 %]	1059
	Mix	0.030	0.032	0.031	6 [9.09 %]	3 [4.55 %]	0	1 [1.52 %]	56 [84.85 %]	66
Date										
53	BIO	0.250	0.094	0.137	2 [4.00 %]	16 [32.00 %]	0	20 [40.00 %]	12 [24.00 %]	50
	Span	0.003	0.019	0.005	293 [84.93 %]	0	2 [0.58 %]	10 [2.90 %]	40 [11.59 %]	345
	Mix	0.250	0.094	0.137	2 [4.00 %]	16 [32.00 %]	0	20 [40.00 %]	12 [24.00 %]	50

Table 3: NER: E3C [SARVI stage 1: Post-training results]

Entities [Support]	Method	P	R	F1	Type 1	Type 2	Type 3	Type 4	Type 5	Total Errors
DISO										
4901	BIO	0.139	<u>0.497</u>	<u>0.217</u>	12678 [83.72 %]	0	21 [0.14 %]	0	2445 [16.21 %]	15144
	Span	<u>0.143</u>	0.448	<u>0.217</u>	10493 [79.49 %]	0	24 [0.18 %]	0	2683 [20.33 %]	13200
	Mix	0.148	0.525	0.231	12476 [84.29 %]	0	22 [0.15 %]	0	2304 [15.57 %]	<u>14802</u>

Table 4: NER: SpCN-Synth [SARVI stage 1: Post-training results]

3 ICD-10-CM results

Here are shown all the results but with detailed-level codes. It is important to note that as SARVI does not output detailed codes in the first stage, the results of the models with HS usage or not, and their

mix with and without confidence, are only displayed in the category-level tables. Also, SpCN-Synth does not have detailed code results, as it is only composed of category-level codes. As with the category-level results, previous state-of-the-art models outperform SARVI in the CodiEsp dataset as they are mainly trained for this specific domain.

ICD-10-CM [Support]	Method	F1 Micro	F1 Macro
CodiEsp			
3665	S2: All	0.615	0.302
	[1]	0.780	–

Table 5: ICD-10-CM: Instance-wise metrics [Details - NOT CodiEsp-like results]

ICD-10-CM [Support]	Method	P	R	F1
CodiEsp				
250	S2: All	0.288	0.450	0.351
	S2: All (Confidence only)	0.323	0.441	0.373
	S2: All (Judger only)	0.288	0.450	0.351
	S2: All (Confidence & Judger)	0.332	0.440	0.378
	[2]	0.866	0.066	0.123
	[2]	0.002	0.897	0.005
	[2]	<u>0.732</u>	<u>0.633</u>	<u>0.679</u>
	[3]	–	–	0.457
	[4]	–	–	0.746
CARES				
403	S2: All	0.024	0.135	0.040
	S2: All (Confidence only)	<u>0.027</u>	0.125	<u>0.045</u>
	S2: All (Judger only)	0.024	<u>0.133</u>	0.041
	S2: All (Confidence & Judger)	0.028	0.124	0.046

Table 6: ICD-10-CM: Document-wise metrics [SARVI: Details - CodiEsp-like results w/ Predicted DISO entities]

Fig. 5 and Fig. 6 show the mean and the interquartile range of the cosine similarity value between the numerical representation of each instance and their closest definition or group in both the no HS and HS methods from the first step of SARVI. The first Fig. corresponds to CodiEsp, and the second to SpCN-Synth. In the no HS method, the correct predictions mostly have a bigger cosine similarity value. However, the first prediction in the HS method is mostly the same cosine value for the correct and incorrect predictions. When mixing, as HS results are used as the base results, in Fig. 5 can be seen that the first step when descending the tree is where most of the errors exist, as the IQR of the incorrect ones is higher than that of the correct. When descending, this does not happen as the distance between both keeps growing. With previously unseen data, it shows that in the HS models significant differences remain at the intermediate levels but disappear at the extremes. This may suggest that training with CodiEsp data enabled the model to learn broader classification patterns, such as letter-level categories. However, as it becomes more specific, the model lacks the fine-grained discriminatory ability learned during training as most of these codes have not been seen.

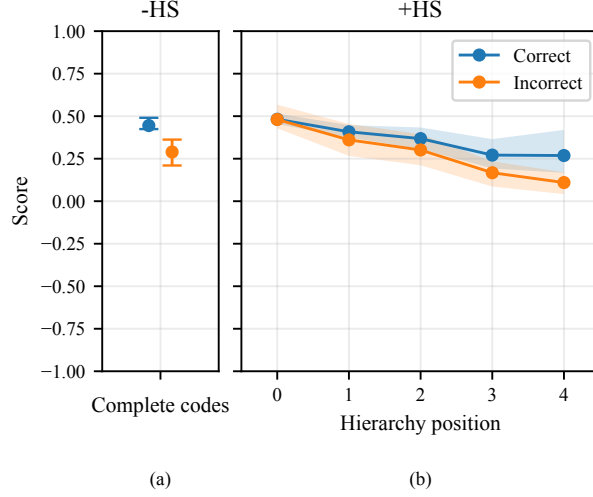


Figure 5: ICD-10-CM: CodiEsp instance-wise cosine similarity values between each numerical representation and their closest definition or group. The mean and IQR are plotted divided by if the prediction is correct or not. Subfigure (a) represents the results with no HS, and (b) with HS.

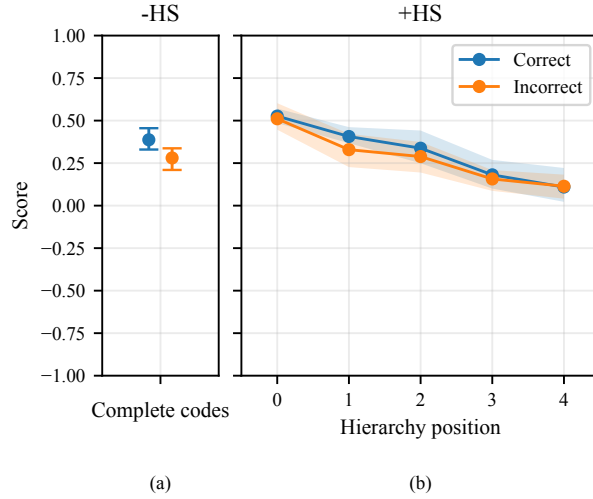


Figure 6: ICD-10-CM: SpCN-Synth instance-wise cosine similarity values between each numerical representation and their closest definition or group. The mean and IQR are plotted divided by if the prediction is correct or not. Subfigure (a) represents the results with no HS, and (b) with HS.

The rest of the results distributions plots made with PCA and UMAP are displayed in Fig. 8, Fig. 7, and Fig. 9. Here, most of the letters have their area of knowledge grouped and there are practically none special outliers correctly or incorrectly predicted.

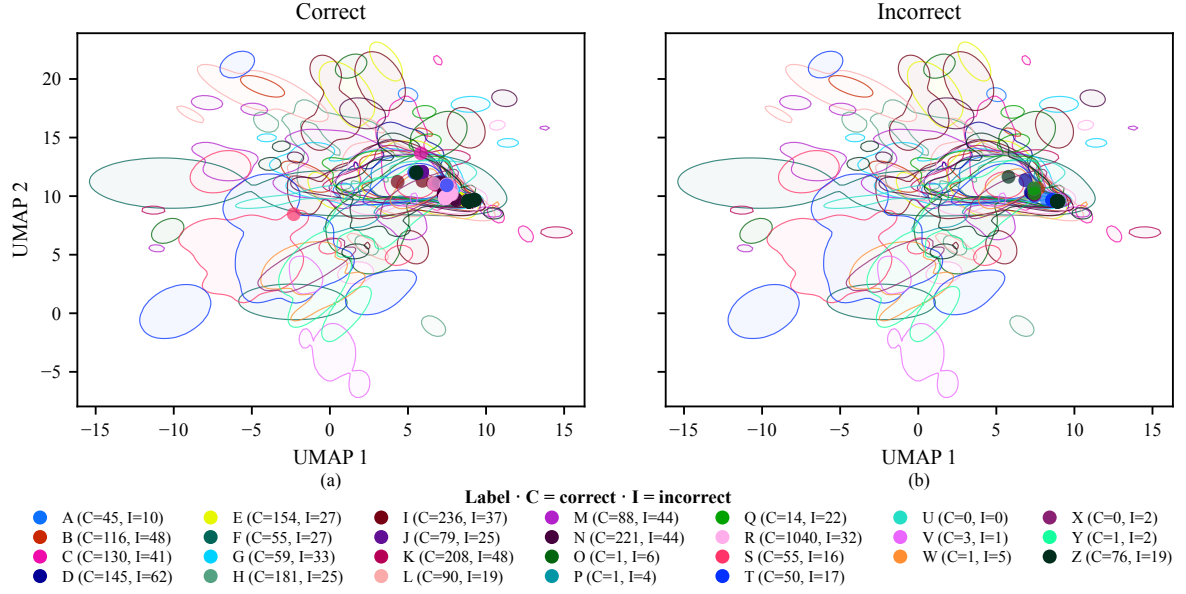


Figure 7: ICD-10-CM: CodiEsp result distribution with no HS. The space occupied by the definitions of all the codes of each letter is represented by the different areas, and each dot is the representation of each instance. Subfigure (a) plots the instances correctly predicted, whereas subfigure (b) plots the incorrect ones.

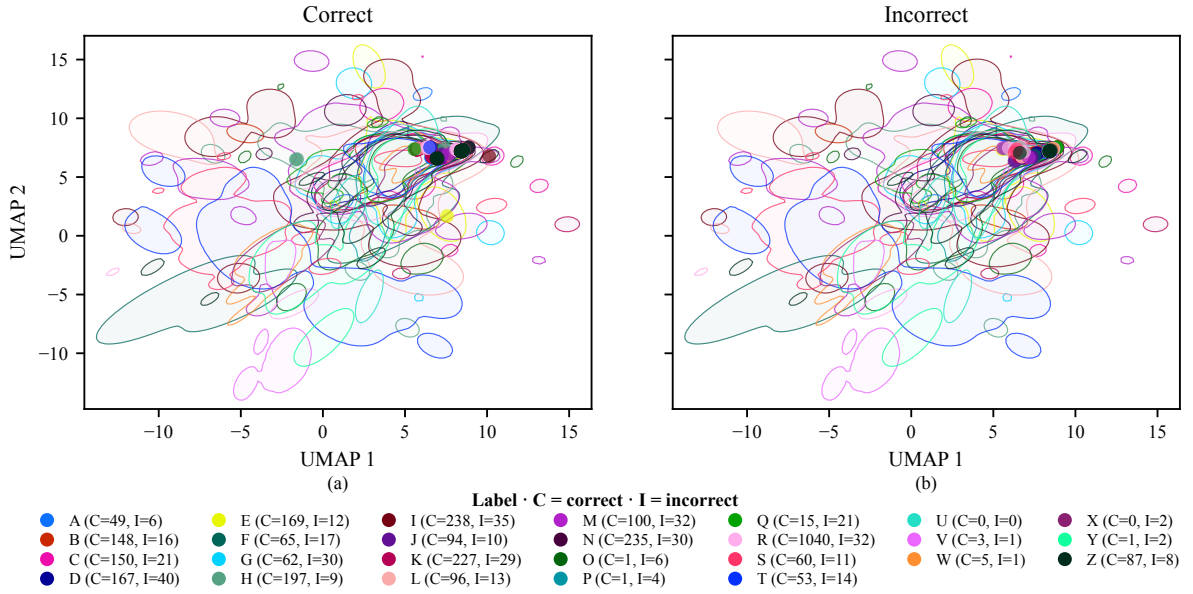


Figure 8: ICD-10-CM: CodiEsp result distribution with HS. The space occupied by the definitions of all the codes of each letter is represented by the different areas, and each dot is the representation of each instance. Subfigure (a) plots the instances correctly predicted, whereas subfigure (b) plots the incorrect ones.

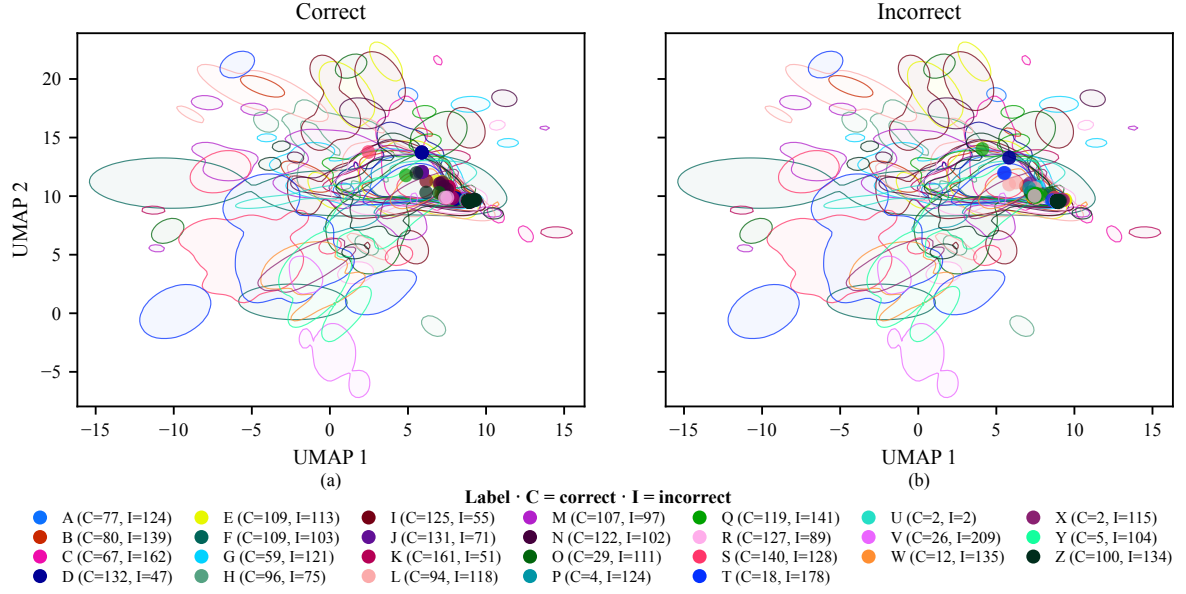


Figure 9: ICD-10-CM: SpCN-Synth result distribution with no HS. The space occupied by the definitions of all the codes of each letter is represented by the different areas, and each dot is the representation of each instance. Subfigure (a) plots the instances correctly predicted, whereas subfigure (b) plots the incorrect ones.

Finally, the results from CodiEsp and SpCN-Synth instance-wise experiments, subdivided by each letter, can be seen in Fig. 10 and Fig. 11. The percentages of code corrections made between stages are also reported. Each subfigure is divided into two rows, the upper being the results from stage 1 and the lower from stage 2. Each one of the three columns depicts the F1-micro scores when evaluating only the letter, the category-code, and the detailed-code, respectively. In addition, the transition between both rows is the percentage of codes that have been corrected in the second stage. The labels (a), (b), (c), ..., represent the results per letter instance-wise, and grey nodes indicate there is no info. Stage 1 results indicate that there is a significant difference in results between datasets. This seems understandable in low-trained letters, but on the contrary same happens. Second, stage 2 corrections are mostly made in the same low-trained letters in CodiEsp, except in SpCN-Synth where practically the correction rate exceeds 50% for most letters. In the first dataset, this corrections do not always assure better scores. Nonetheless, in the other one, all changes drastically improve the scores per letter. It can be concluded that the second stage, as it has not been trained on any specific data, it is more general and the corrections made on the specific trained distribution of the first stage are not the best, but on the unseen data they are a great improvement.

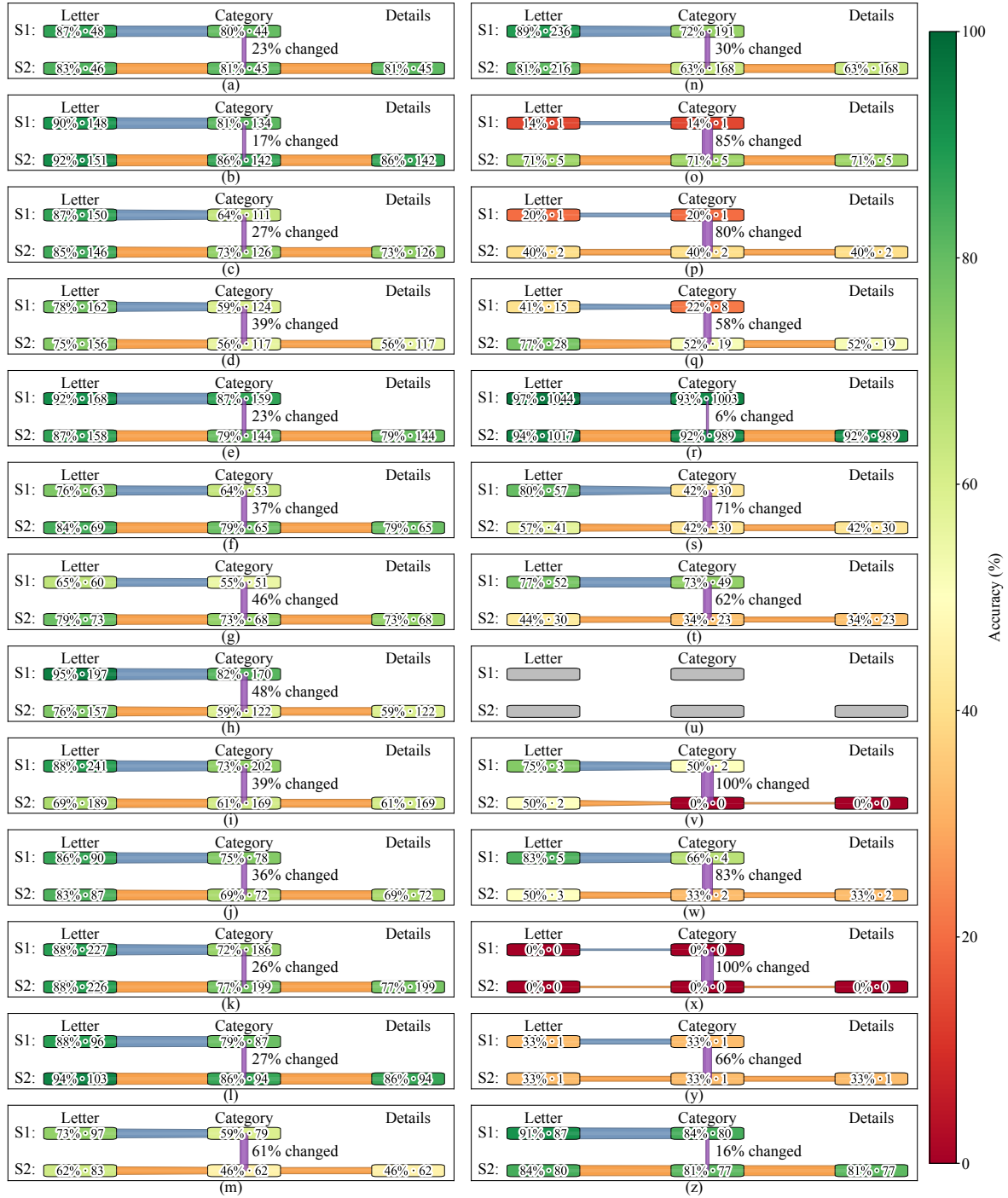
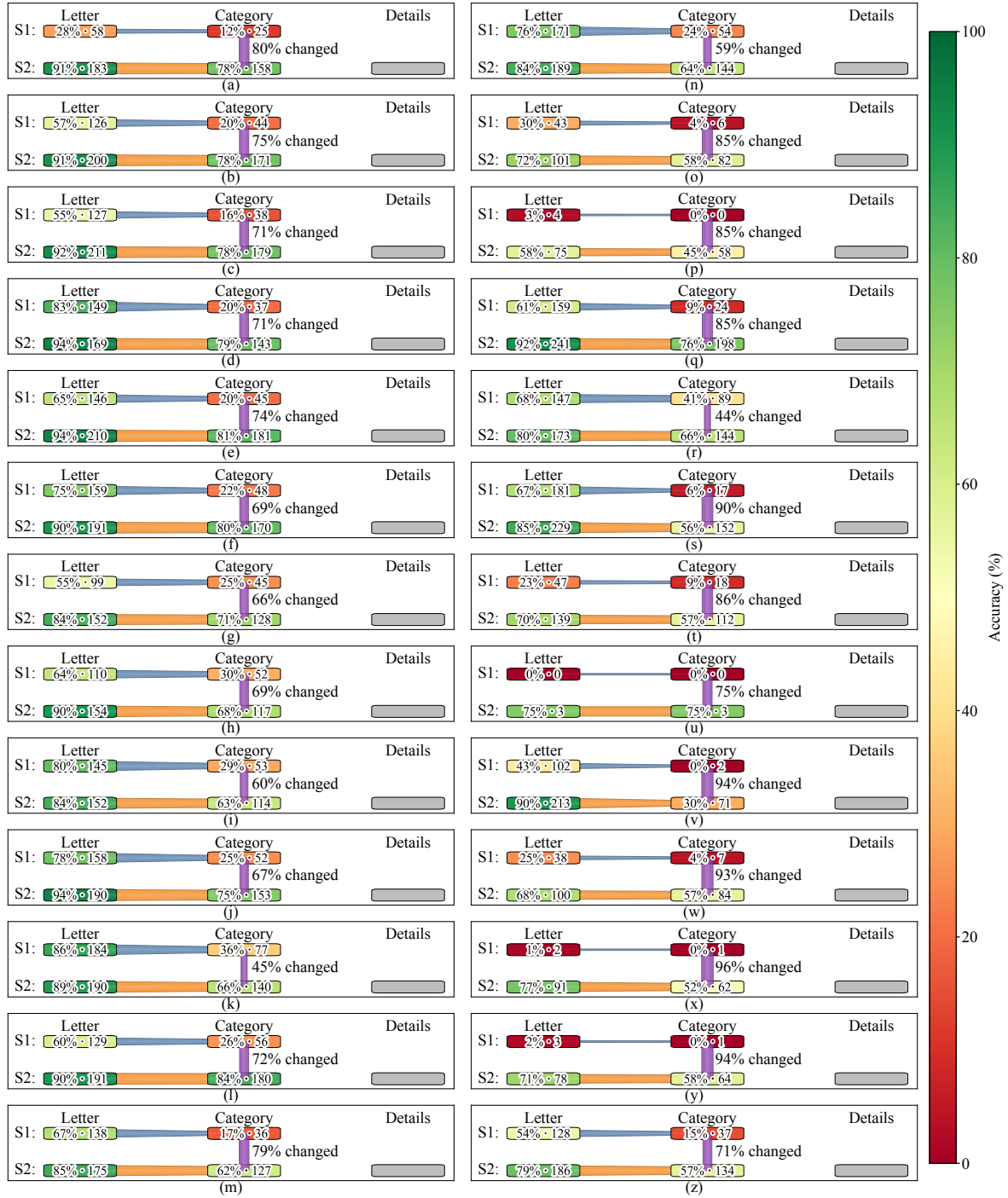


Figure 10: ICD-10-CM: CodiEsp result per letters.



Label format: Accuracy (%) · N° of correct instances

Figure 11: ICD-10-CM: SpCN-Synth result per letters.

References

References

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